

PhD researcher – Center for Molecular Modeling, UGent

I am an engineer trained in applied physics and materials modeling
looking for career paths in research & development, materials science
or computational physics/chemistry.

MSc Engineering Physics (2019)

PhD Engineering Physics (june 2026)

PROFESSIONAL EXPERIENCE

Researcher

Center for Molecular Modeling

- Combining knowledge from materials science, physics and (quantum) chemistry aimed at understanding industrial materials at the atomic level.
- Building workflows that characterise material structure and behavior from experimental sample to atomic & electronic structure through nuclear magnetic resonance, X-ray diffraction & molecular modeling.
- Close collaboration with experimental partners and industry:

Umicore Automotive Catalysts (2019-2021)

Centre for Ordered Materials, Organometallics and Catalysis (UGent, 2021-2024)

Centre for Surface Chemistry & Catalysis (KU Leuven, 2019-now)

Teaching Assistant

Department of Applied Physics

Focus on physics and mathematics education.

Affiliated with courses in Bachelor of Engineering: Molecular Structure, Physics II, Basic Mathematics.

Supervised 2 MSc thesis students in molecular modelling.

EDUCATION

Doctor of Engineering Physics

UGent

Topic: Characterisation of complex materials through spectroscopic modelling of nuclear magnetic resonance properties. Graduation: june 2026.

Promotors: prof. dr. Veronique Van Speybroeck (UGent) & dr. Eric Breynaert (KU Leuven)

Master of Science in Engineering Physics -- cum laude

UGent

Dissertation: From a dense to a sponge-like structure at high pressure: unravelling the counterintuitive behaviour of zinc cyanide.

SKILLS

Solid-state physics, multiscale atomic & electronic structure modelling, experimental model validation

Engineering: large-scale simulations management, automated screening, design of experiments to characterise industrial materials, statistical analysis of modelling results

Scientific reporting and communication in various contexts (reporting to academic and industrial partners, faculty and students)

Grant writing for compute infrastructure, software testing as pilot user for compute infrastructure

Quantum chemistry software: VASP, CP2K, ORCA, Gaussian, QuantumESPRESSO

Workflow & visualisation: Python & bash scripting, ASE, Pymatgen, VESTA, VMD

Languages: Dutch (native), English (full), French & German (limited)

SELECTED PUBLICATIONS

1. *Computational Protocol for the Spectral Assignment of NMR Resonances in Covalent Organic Frameworks*
Vanlommel et al., Journal of Chemical Theory and Computation, 2024 (A1 paper).

2. *How water and Ion Mobility Affect the NMR fingerprints of the Hydrated JBW zeolite: A Combined Computational-Experimental Investigation*

Vanlommel et al., Chemistry – A European Journal, 2022 (A1 paper).

3. *Hindered Dynamic Diversity: NMR investigation of Cation Species in Zeolite Growth Systems*
Vanlommel et al. 2026 (submitted).

4. *Deciphering Framework Aluminium in Zeolites via a Computational-ssNMR Strategy: The Case of Chabazite*
Vanlommel et al. 2026 (submitted).

5. *Engineering of phenylpyridine- and bipyridine-based covalent organic frameworks for photocatalytic tandem aerobic oxidation/Povarov cyclization*

Debruyne et al., ACS Applied Materials & Interfaces, 2023 (A1 paper).

GRANTS, AWARDS

Various grants on high-performance computing systems LUMI (Tier-0), Hortense/BrENIAC (Tier-1), totalling 40M+ core-hours of compute time

Young Scientist Award – International Union of Crystallography, 2024.

Talks

How Water & Ion Mobility Affect the NMR Fingerprints of the Hydrated JBW Zeolite

Siebe Vanlommel, Eric Breynaert, Veronique Van Speybroeck

Winterschool for Theoretical Chemistry and Spectroscopy, 2022 (Han-sur-Lesse, Belgium)

Computational NMR as an asset to structural investigation of complex materials

Siebe Vanlommel, Eric Breynaert, Veronique Van Speybroeck

Smarter7, 2023 (Leuven, Belgium)

Challenges in simulating aluminosilicate zeolites and connecting NMR theory to experiment

Siebe Vanlommel, C. Vinod Chandran, Sambhu Radhakrishnan, Eric Breynaert, Veronique Van Speybroeck

Smarter8, 2024 (Aveiro, Portugal)

Pushing the limits of computational chemistry: the phase diagram of water

Massimo Bocus, Tom Braeckevelt, Pieter Dobbelaere, Arnout Maet, Wim Temmerman, Sander

Vandenhaute, Siebe Vanlommel, Jelle Vekeman, Veronique Van Speybroeck

EuroHPC User Days, 2025 (Copenhagen, Denmark)

Active Sites in MAPO-18 zeotypes for CO₂ hydrogenation: an NMR study

Wim Temmerman, Siebe Vanlommel, Jose Nuno Almeida, Sang-HoChung, Christophe Coudercy, Bjørnar

Arstad, Javier Ruiz-Martínez, Unni Olsbye, Veronique Van Speybroeck

NCCC XXVII, 2026 (Noordwijkerhout, The Netherlands)

Posters

Computational Investigation into the nuclear magnetic resonance properties of zeolite ferrierite

Siebe Vanlommel, Alexander E.J. Hoffman, Veronique Van Speybroeck

MolSim, 2021 (online)

Characterising the effect of hydration on nuclear magnetic resonance properties in zeolites

Siebe Vanlommel, Alexander E.J. Hoffman, Sam Smet, Sambhu Radhakrishnan, Karel Asselman, C. Vinod

Chandran, Eric Breynaert, Christine E.A. Kirschhock, Johan A. Martens, Veronique Van Speybroeck

FEZA, 2021 (online) & Water in Zeolites Workshop, 2021 (Czech Republic) & Smarter7, 2023 (Belgium)

Challenges in simulating aluminosilicate zeolites and connecting NMR theory to experiment

Siebe Vanlommel, C. Vinod Chandran, Sambhu Radhakrishnan, Eric Breynaert, Veronique Van Speybroeck

Smarter8, 2024 (Portugal)